

Formal Methods in Software Engg.

Assignment 6 (AFL) - Problem 1

Part b)

After running AFL for over 20 cycles, there were no test input files generated using bit flips. The entire structure of the output directory is as follows:

```
.
├── crashes
├── fuzz_bitmap
├── fuzzer_stats
├── hangs
├── id:000000,src:000016,op:havoc,rep:16
├── plot_data
└── queue
    ├── id:000000,orig:seed
    ├── id:000001,src:000000,op:int8,pos:1,val:+0
    ├── id:000002,src:000000,op:havoc,rep:4
    ├── id:000003,src:000000,op:havoc,rep:8
    ├── id:000004,src:000000,op:havoc,rep:64
    ├── id:000005,src:000000,op:havoc,rep:16,+cov
    ├── id:000006,src:000000,op:havoc,rep:128,+cov
    ├── id:000007,src:000000,op:havoc,rep:16
    ├── id:000008,src:000000,op:havoc,rep:32
    ├── id:000009,src:000000,op:havoc,rep:128
    ├── id:000010,src:000000,op:havoc,rep:64
    ├── id:000011,src:000000,op:havoc,rep:8
    ├── id:000012,src:000000,op:havoc,rep:32
    ├── id:000013,src:000011,op:havoc,rep:2
    ├── id:000014,src:000011,op:havoc,rep:2
    ├── id:000015,src:000012,op:havoc,rep:4
    ├── id:000016,src:000012,op:havoc,rep:64
    ├── id:000017,src:000014+000015,op:splice,rep:4
    ├── id:000018,src:000016,op:havoc,rep:4
    ├── id:000019,src:000016,op:havoc,rep:4
    ├── id:000020,src:000016,op:havoc,rep:16
    └── id:000021,src:000015,op:havoc,rep:16
```

4 directories, 26 files

The AFL terminal output is as follows:

american fuzzy lop 2.35b (for-class-while)

process timing		overall results	
run time : 0 days, 0 hrs, 0 min, 20 sec		cycles done : 26	
last new path : 0 days, 0 hrs, 0 min, 10 sec		total paths : 22	
last uniq crash : none seen yet		uniq crashes : 0	
last uniq hang : 0 days, 0 hrs, 0 min, 13 sec		uniq hangs : 1	
cycle progress		map coverage	
now processing : 5* (22.73%)		map density : 0.02% / 0.02%	
paths timed out : 0 (0.00%)		count coverage : 5.33 bits/tuple	
stage progress		findings in depth	
now trying : havoc		favored paths : 2 (9.09%)	
stage execs : 221/256 (86.33%)		new edges on : 3 (13.64%)	
total execs : 231k		total crashes : 0 (0 unique)	
exec speed : 11.1k/sec		total hangs : 1 (1 unique)	
fuzzing strategy yields		path geometry	
bit flips : 0/3920, 0/3898, 0/3854		levels : 4	
byte flips : 0/490, 0/425, 0/395		pending : 0	
arithmetics : 0/24.7k, 0/5557, 0/1151		pend fav : 0	
known ints : 1/2228, 0/11.3k, 0/17.3k		own finds : 21	
dictionary : 0/0, 0/0, 0/0		imported : n/a	
havoc : 19/133k, 1/23.1k		stability : 100.00%	
trim : 30.58%/182, 8.45%			
^C		[cpu000: 35%]	

+++ Testing aborted by user +++

[+] We're done here. Have a nice day!